Department of Computer Science and Business Systems
Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: B.Tech., VI Semester & CSBS
Course Code & Title: CCS356 & Object Oriented Software Engineering
Name of the Faculty member (s): Ms.M. Preethi Ram, AP/CSBS

Innovative Practice Description

- **Unit / Topic: Unit 5/ Deployment Pipeline**
- **Course Outcome:** CO5
- **Activity Chosen:** Think –Pair Share

- **Justification:**

  The Deployment Pipeline involves multiple stages such as building, testing, and deploying software, which require a clear understanding of sequence, tools, and integration. The **Think-Pair-Share** activity encourages students to individually reflect on each stage of the pipeline, pair up to discuss and clarify their understanding, and finally share insights with the class. This collaborative approach enhances critical thinking and deepens comprehension of how continuous integration and deployment processes work in real-world scenarios. It also allows students to learn from diverse perspectives, correct misconceptions, and solidify their understanding of complex concepts through peer discussion.

- **Time Allotted for the Activity:** 35 Minutes

- **Details of the Implementation:**

  A Collaborative, active learning strategy, in which students work on a question posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share).

  • **Think –** Write- Students were provided with questions on Deployment Pipeline. The student has to think individually and write the answer in a paper.
  • **Pair-** After writing the answers, the students were allowed to discuss with neighbor and justify the resultant answer to each other. The pair with neighbor was shown in Fig.2
  • **Share -** Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team and it is shown in the Fig 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO11	PSO1	PSO2
CO 5	2	2	2	1	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO11**	**PSO1**	**PSO2**
	2	2	2	1	2	2
Justification for correlation	Students apply knowledge of software engineering fundamentals and modern development practices to understand the structure and functioning of the deployment pipeline.	Students analyze issues that arise in different stages of the pipeline (e.g., build failures, test coverage, deployment errors) and identify appropriate solutions.	Students design efficient, secure, and automated deployment pipelines that meet real-world software development requirements.	Students understand how deployment automation reduces time-to-market, optimizes resources, and fits into the broader project management lifecycle.	Students apply tools and techniques from Cloud Computing, Cybersecurity, and DevOps to develop reliable and secure deployment pipelines that support scalable and contemporary software systems.	By working on deployment strategies and DevOps tools, students gain practical skills aligned with industry needs, enhancing their software development and system integration proficiency.

- **Images / Screenshot of the practice:**

Fig.1. Saravanan presented the topic to the students

Fig.2. Haritha and Kousalya as a team shared their ideas to her classmates

Reflective Critiques

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Some of the students write answers for the questions easily and share their ideas with others.
- ❖ Some of them need more explanation and examples to get clarity.
- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and get more confidence, clear understanding in this topic when discussing with their peers.

- ❖ **Challenges faced in implementation:**
  - The students who didn't understand the concept clearly is not able to complete it within time.
  - Making all the students to involve and present actively in this activity is a challenging.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share_PJT.pdf
- ❖ https://www.youtube.com/watch?v=1yRJf8j8xVU